Making a Markov Chain from a given text:

1) Draw a "[Start"] box on the left side of your piece of paper and to the right of it draw a box containing the first word of the text. Draw an arrow linking "[Start]" to the first word of the text. Think of that first word as the "Current word" and underline it in the text to keep track of where you are.

2) If the underlined "Current word" is the last word in the text and it already has an arrow to a "[Stop]" box, stop the assignment. If the underlined "Current word" is the last word in the text and it does not already have an arrow to a "[Stop]" box, draw a "[Stop]" box to the right of the box for "Current word" and link the "Current word" box to the [Stop] box with an arrow and then stop this assignment.

3) If the underlined "Current word" is not the last word in the text, count how many times "Current word" appears in the text (including the underlined occurrence) and write that number down underneath a forward-slash division sign, as in "/3" if "Current word" appears in the text three times.

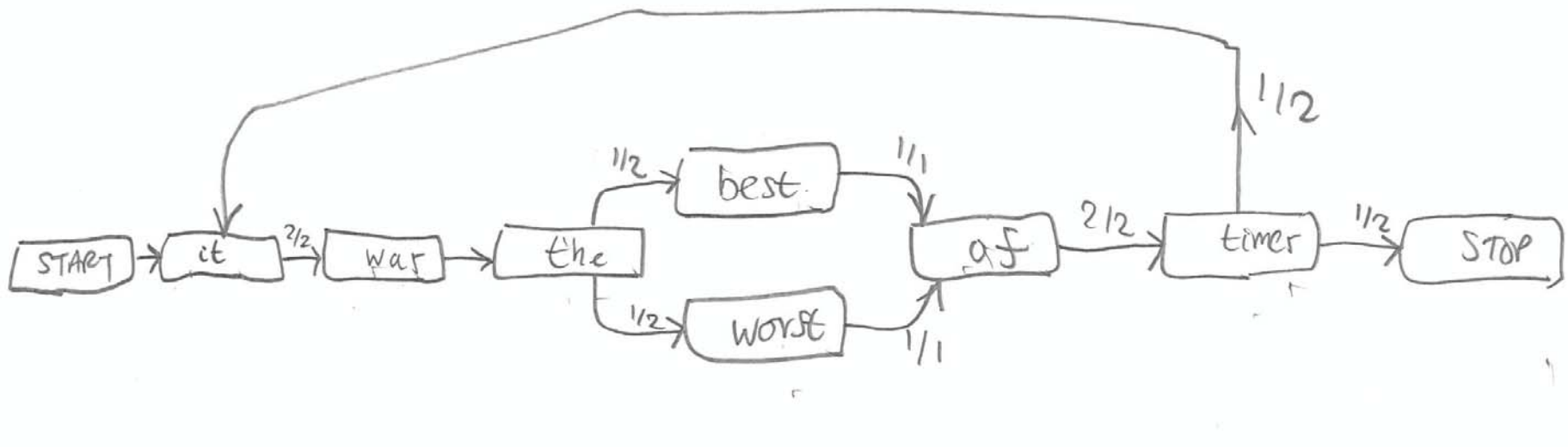
4) On a spare piece of paper, make a "Comes next" list for "Current word" as follows. For each occurrence of "Current word" in the text, including the underlined one, write down the single word that immediately follows it. Beside each of these words that "Comes next" write how many times it "Comes next" after the "Current word" in the text and follow that count with the divisor from Step (3). Thus, if "Current word" appears in the text three times there must be three words that follow "Current word", but they are not necessarily three different words. If "Current word" is followed by the word "the" two times and is followed by the word "on" one time, write down "the 2/3" and "on 1/3" in the "Comes next" list. If one of the occurrences of "Current word" in the text is the last word in the text, add the item "[Stop]" to the "Comes next" list and add its fraction (which will be 1 over the number of occurrences of "Current word" in the text).

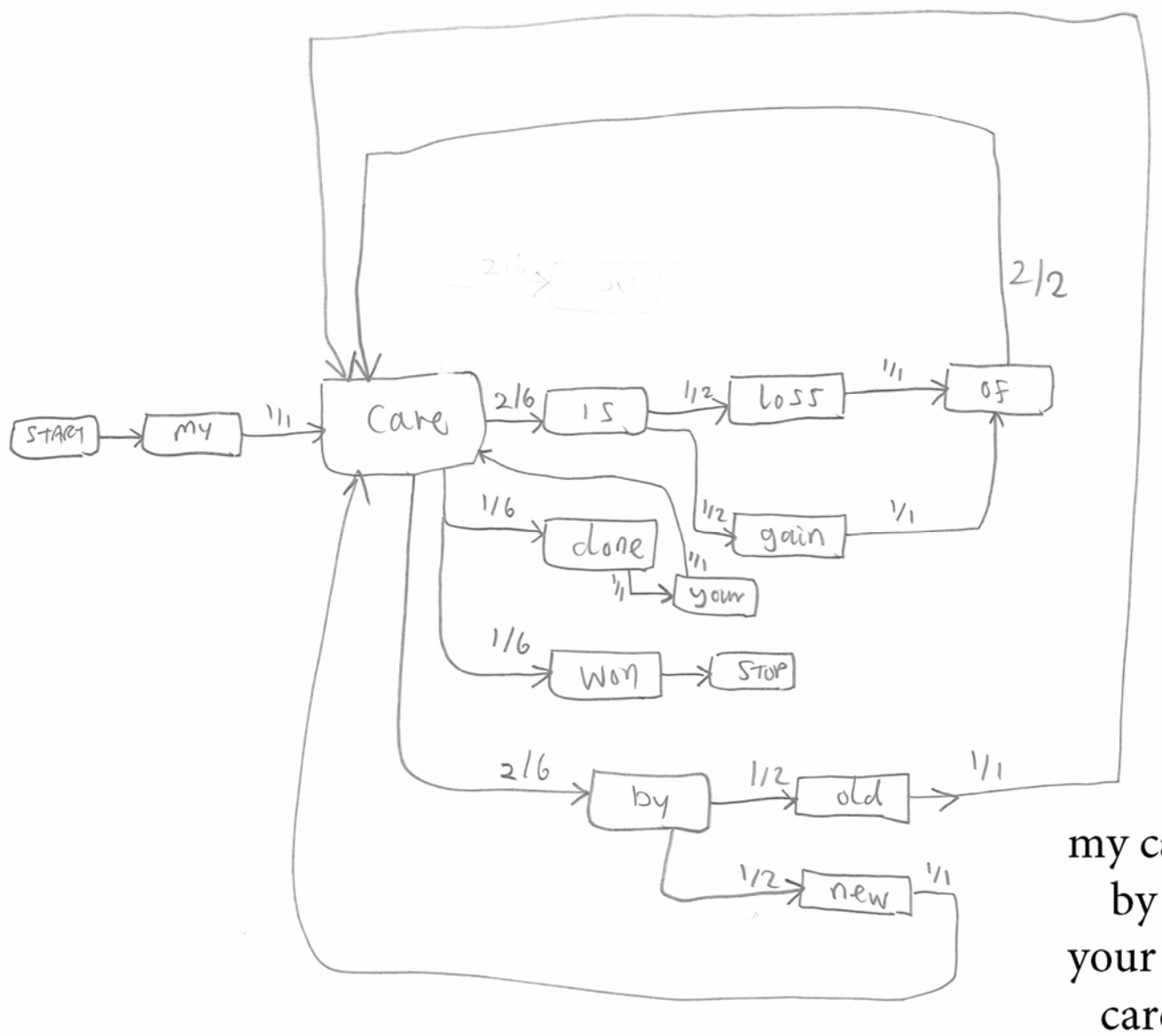
5) Returning to your Markov Chain, draw a box to the right of the "Current word" box for each of the words (or the "[Stop]" token) in the "Comes next" list you made in Step (4) that you don't already have a box for, then put the word (or the "[Stop]" token) from the "Comes next" list inside the box and draw an arrow from the "Current word" box to each of these new boxes. If one of the words (or the "[Stop]" token) in the "Comes next" list made in Step (4) already has a box drawn in a previous step, don't draw a new box but instead draw the arrow from the "Current word" box to the existing box. Beside each arrow put the associated "weight" from the "Comes next" list made in Step (4). Thus, if "Current word" is followed by the word "the" two times and is followed by the word "on" one time, write beside the arrow from the "Current word" box to the "the" box the fraction "2/3" and write beside the arrow from the "Current word" box to the "on" box the fraction "1/3".

6) Cross "Current word" off from the text. The next word in the text is your new "Current word", so underlined it. You must already have a box for this new "Current word" since you just drew it in Step (5) or in a previous iteration of that step.

7) Go to Step (2)

it was the best of times it was the worst of times





my care is lost of care
by old care done
your care is gain of
care by new care won